

**B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER
EXAMINATION 2026**

DIGITAL SIGNAL PROCESSING

Full Marks 100

Time: Three hours

(50 marks for each part)

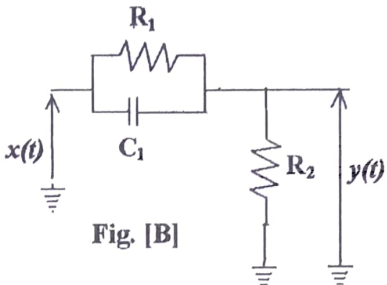
Use a separate Answer-Script for each part

Question No.	PART-I	Marks
	Answer any THREE questions Two marks reserved for well organized answers and neatness	
1. (a)	A signal $f(t) = 2 \cos^4(20\pi t)$, where t is in ms, is uniformly sampled. (i) Determine the Nyquist rate for the signal. (ii) If the sampling frequency is $0.7 \times$ the Nyquist rate, and the sampled signal is then converted back to analog signal by ideal low-pass filtering with a cutoff frequency equal to half the sampling frequency, determine the frequencies of the harmonic components present in the reconstructed analog signal. Derive only the specific expression used in Part (ii).	10
1 (b)	The Z-transform of a sequence $f[n]$ is $F(z) = \log\left(1 - \frac{1}{3}z^{-1}\right); \quad z > \frac{1}{3}.$ Verify whether or not $f[n] = -\frac{1}{n}\left(\frac{1}{3}\right)^n u[n-1].$ If not, determine the exact expression for $f[n]$. OR	6
1. (b)	Verify the correctness of the following expressions for Z-Transforms. (i) $\mathcal{Z}[u_{-n}] = \frac{1}{1-z}$ (ii) $\mathcal{Z}[u_n - u_{n-N}] = \frac{z^N - 1}{z^{N-1}(z-1)}$ If the expression(s) are incorrect, obtain the correct expressions for the Z-transforms. Also determine the ROCs.	6

3, 2, 6, 7

[Turn over

Question No.	PART I	Marks
2(a)	Determine the impulse response of the DTLTI system shown in Fig. [A], without using any transform method. Fig. [A] The impulse responses of the subsystems are $g_1[n] = \{-1, 3, 5, -2, 1\}, \quad g_2[n] = \{0.5, 1, 6, 8, -1, 2\}$ $h_1[n] = \{2, -0.5, -1, 7, -0.5, -1\}, \quad h_2[n] = \{-1.5, 2, -6, -1, 4, 1\}$ OR 2. (a) The transfer function of a DTLTI system is $H(z) = \frac{z^4 + z^3}{z^2 - 0.75z + 0.125}$ Determine all the possible ROCs of $H(z)$ and for each of them, find the impulse response of the system in closed form. Also point out any important property of the system revealed by the impulse responses/ ROCs. 2. (b) How does the $j\omega$ axis in the s-plane map on to the z-plane? Explain. 3. (a) In the continuous-time filter circuit shown in Fig. [B], $x(t)$ is the input and $y(t)$ is the output. The components have the following values. $R_1 = 10 \text{ k}\Omega, C_1 = 1 \text{ }\mu\text{F}, \text{ and } R_2 = 100 \text{ k}\Omega.$	10 10 6

Question No.	PART I	Marks
	Discretize the filter transfer function to obtain the equivalent digital filter transfer function. Use bilinear transformation in conjunction with frequency prewarping. Consider a sampling rate of 1.6 Hz. Also give the difference equation relating the output and the input sequences of the digital filter.  Fig. [B]	10
3.(b)	Explain the problem of 'Characteristic Aliasing' inherent in the Impulse Invariant Transformation Technique. Under what conditions is the effect of this problem be minimal?	6
4. (a)	The output $y[n]$ and the input $x[n]$ of a DTLTI system are related through the difference equation $y[n] = x[n] + 3x[n-1] + 2x[n-2] - \frac{3}{8}y[n-1] + \frac{3}{32}y[n-2] + \frac{1}{64}y[n-3]$ Derive and draw the Direct form-I, the Direct form-II and the cascade realization (using 1st order subsystems with direct form-II structure) of the system. Give all the relevant sets of difference equations.	10
4.(b)	A causal DTLTI system is represented by the following difference equation relating the output $y[n]$ and the input $x[n]$. $y[n] = b_0 x[n] + a_1 y[n-1]$ Show that the system is an IIR system. Determine the effect of the coefficient values on the stability of the system.	6

Question No.	PART- I	Marks
5. ✓(a) (b) ✓(c) (d)	Write short notes on any two of the following: Discrete-time integrators. FIR and IIR DTLTI systems. Representing uniform sampling as impulse modulation. Stability of DTLTI Systems.	8+8

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No. of Questions	PART II	Marks
	<i>Answer all the questions.</i>	
1 ✓	Show in detail how can the Fourier series of a periodic signal be computed in terms of sine coefficients and cosine coefficients. How are the amplitude and phase of the nth harmonic related to these sine and cosine coefficients? (CO1-K1) OR a) Using the DFT coefficients $X_n = [20 \quad -4 + 4i \quad -4 \quad -4 - 4i]$, compute the inverse discrete Fourier transform to obtain the time domain sequence x_k, $k=0,1,2,3$. 06 (b) What is circular symmetry of a sequence? What is the importance of "modulo" operation in it? Explain with a suitable example. (CO1-K1) 04	
2. (a)	Derive the symmetry property of the impulse sequence of a causal FIR filter. Then derive the frequency response of this causal FIR digital filter. (CO2-K2) 09 OR Give a step-by-step procedure of employing two-dimensional discrete convolution summation for offline analysis of two-dimensional data using FIR digital filters. (CO2-K2)	
(b)	Show that the direct realization of an M-th order linear phase FIR digital filter can be achieved using $\left(\frac{M+1}{2}\right)$ number of multiplications, when M is odd. What will be the number of additions needed for realizing this filter? (CO2-K2) 07 OR Show that, in rectangular window based design of FIR filters, the frequency response of the filter designed can be expressed in form of a circular complex convolution integral. What is Gibbs phenomenon? How can it be reduced? (CO2-K2)	

$$f_n = \frac{a_n - j b_n}{2}$$

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No. of Questions	PART II	Marks
3.	Write a short note on any one of the following: (CO3-K3) (i) Splitting an $(N/2)$-point DFT into two $(N/4)$-point DFTs. (ii) Computation of 4-point FFT. ✓	08
4.	A 5-tap FIR causal digital filter with <u>stepped characteristic</u> is designed with its frequency response given as: For $-\frac{\omega_s}{2} \leq \omega \leq \frac{\omega_s}{2}$, $H(\omega) = 1.1e^{-j\omega\tau(M-1)/2}, \quad \text{for } \omega \leq \omega_1 \quad \checkmark$ $= 0.9e^{-j\omega\tau(M-1)/2}, \quad \text{for } \omega_1 < \omega \leq \omega_2 \quad \checkmark$ $= 0, \quad \text{otherwise}$ where all symbols have usual meaning. Here, $f_1=200$ Hz, $f_2=400$ Hz, and sampling frequency $f_s=1000$ Hz. The filter coefficients are smoothened using Hann window. Realize the filter and draw its schematic form. (CO4-K3) OR Derive the frequency response of a non-causal Blackman window, whose window sequence is given as: $w_n = \begin{cases} 0.42 + 0.5\cos\left(\frac{2\pi n}{M-1}\right) + 0.08\cos\left(\frac{4\pi n}{M-1}\right), & \text{for } n \leq \frac{M-1}{2} \\ 0, & \text{otherwise} \end{cases}$ Comment on its approximate width of the main lobe. (CO4-K3)	08

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No. of Questions	PART II	Marks
5.	Justify or correct <u>any two</u> of the following statements with suitable reasons/derivations, in brief. (CO5-K4) i) The DFT coefficients (X_n) for an N-point periodic discrete sequence can be obtained from the corresponding a_n and b_n coefficients by employing the relation: $X_n = \left(\frac{N}{2}\right)(a_n + jb_n)$, for $n = 0, \pm 1, \pm 2, \pm 3, \dots$ ii) Radix-2 decimation-in-frequency FFT algorithm can incorporate an "in-place" feature because of the structure of each butterfly employed. iii) Both phase delay and group delay can be equally employed in all conditions to test whether a digital filter permits distortion-less transmission of signal or not.	04×02 =08